Oleksii Odarchuk

Backend Developer (Go) · Creator of the Piton language

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Performance-obsessed engineer with a product mindset. 1st place at BEST Hackathon 2026, 3rd at Mate Hackathon, invited to DOU Day 2026 as a Student with High Potential. Builds real products fast — from hackathon MVPs to production Go services.

🚀 WINS & HACKATHONS

🏆 1st Place — Lead Backend & Sensor Fusion

April 18–19, 2026

BEST Hackathon 2026 — GPS-less Drone Trajectory

- Algorithm to **reconstruct a UAV's trajectory without GPS** — from IMU, barometer and other inertial-sensor logs. Sensor fusion in **Go + C**: Go for the pipeline & integration, C inline for the hot numerical kernels.
- **1st place among top teams** — judges singled out the architectural resilience: modular split telemetry / integration / simulation let us swap datasets without rewriting.

🥉 3rd Place — Backend Engineer

March 14, 2026

Mate Hackathon (50 participants, 10 teams)

- Designed and built a Go backend from scratch in **7 hours** for an AI-powered MarTech tool that automates competitor-ad analysis.
- Dual AI pipeline: GPT-4o for marketing trigger analysis + Nano Banano 2 (fal.ai) for creative generation; backend routed all AI calls to cloud APIs, sustaining throughput under hackathon load.
- Beat teams with 5+ years of experience by focusing on a stable MVP — **3rd of 10 teams**.

Security Researcher → Job Offer

2026

College Blockchain System "Student Hryvnia"

- Discovered and exploited a critical vulnerability: frontend capped transaction amounts, **backend had no such check** — enabling arbitrary negative-value transfers. Reported responsibly; received a job offer from administration.
- Reverse-engineered the official JS miner through Webpack bundles — then built a faster Go replacement with a goroutine worker pool (SHMiner — p. 2).

🎟️ DOU Day 2026 — Student with High Potential

Personally invited by the organisers with a free ticket — a dedicated track for promising student developers.

🛠️ TECH STACK

Languages: Go Python C TypeScript

Backend: Gin Fiber REST WebSockets Goroutines JWT OAuth 2.0

Databases: PostgreSQL Redis ClickHouse

Frontend: Svelte Wails Tailwind

AI / APIs: GPT-4o Fal.AI

DevOps: Docker Linux (Arch) Git TinyGo Bruno Typst

🎓 EDUCATION & LANGUAGES

Network Technologies — IT College NULP, 2nd year
2024 — present

IT Class — Science Lyceum, Rivne
2022 — 2024

Ukrainian — native · **English** — B1

💻 PET PROJECTS

🐍 Piton — My own programming language

github → playground → docs →

Ukrainian transliterated language + interpreter + flowchart generator

Interpreted language with an authentic Ukrainian syntax (in transliteration). Written from scratch in **Go without yacc/bison**: recursive descent parser, tree-walking evaluator, native SVG flowchart generation (D2), Turing-complete. Web playground: interpreter compiled to **WebAssembly via TinyGo** (220 KB vs. 24 MB with stock Go) + a separate Go-WASM module for D2 visualization (lazy-loaded).

Go · TinyGo · WebAssembly · AST · D2 / SVG · Lexer + Parser · Nix Flakes

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OPEN SOURCE SDK

go-monobank-sdk — Go SDK for every monobank API

github →

Full coverage of **personal**, **provider** (with monoKEП), **business** for legal entities, **acquiring** (subscriptions, monopay button, split-receivers, T2P) and **Installment Purchases**. ECDSA webhook verification with auto-rotating keys, drop-in `http.Handler`, typed money/currency/MCC, HMAC-SHA256 for installment callbacks, a fake monobank server for tests, and a separate OpenTelemetry sub-module.

Go · ECDSA · HMAC-SHA256 · OpenTelemetry · iter.Seq2

SDK CONSUMER

monokasa — Telegram Ticket-Selling Bot

github →

Sells tickets to a single show through a monobank jar and issues PDFs with QR codes scanned at the venue entrance. Live consumer of my own **go-monobank-sdk**: webhook auto-registers at startup, reservations get a 15-minute hold, and show-time reminders fire at-most-once even across restarts.

Go · SQLite · telebot · PDF + QR · go-monobank-sdk

1000× FASTER

SHMiner — Desktop Crypto Miner

github →

Cross-platform desktop miner built with **Wails (Go + Svelte/TypeScript)**. Rewrote SHA-256 hashing from JS to Go with a goroutine worker pool — achieved **1000× faster hashrate** than the official client on identical hardware. AES-encrypted wallet storage, real-time WebSocket dashboard, multi-wallet, zen mode.

Go · Svelte · TypeScript · Wails · WebSockets · Goroutines

150+ USERS

AbitAssistant Bot

github →

Telegram bot helping Ukrainian students track university admission rankings. Reverse-engineered closed third-party APIs to fetch live competition data. **150+ active users in first year**, marketing campaign planned for 2026.

Python · Aiogram · PostgreSQL · BeautifulSoup · Pandas · Docker

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LinkShortener Bot

github →

Telegram URL shortener with QR generation and geo analytics. PostgreSQL for storage, Redis for redirect caching, ClickHouse for real-time click analytics (country, city, platform). Docker, GeoIP2.

Go · PostgreSQL · Redis · ClickHouse · Docker · Telegram API

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rombik — Flowchart Generator

github →

Free tool that instantly generates state-standard (DSTU) flowcharts from Python code, saving hours of manual drawing for academic papers. Supports conditions, loops, exceptions, and functions. Automatically splits giant flowcharts across pages using connectors. Features a built-in visual editor (Svelte 5) and 1-click export to vector formats, Typst, and PDF.

Go · WebAssembly · Svelte 5 · Python Parser